QR BASED LIBRARY BOOK RECOMMENDER SYSTEM USING CONTENT-BASED FILTERING WITH HIDDEN MARKOV MODEL

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

A library is a physical building or room containing a collection of accessible books and resources. It contains books and resources on various genres and topics. Libraries are essential for the healthy growth of society. To meet the people's educational needs, libraries provide invaluable services. People who use libraries for research and to improve their job skills also contribute to our country's economic growth. They are crucial to the overall educational development of individuals and the community.

Today, the amount of information, especially on the Internet, is expanding at a rapid rate. Finding pertinent information becomes increasingly difficult. Recommendation systems aim to resolve such issues. With their assistance, one can quickly access relevant information without manually searching the Internet. Therefore, many websites use recommendation systems to promote and sell their products today. Numerous products, including music, movies, articles, etc., can be recommended to customers based on their profiles in online stores or even social networks, browsing history such as visited links, browsing activity such as the number of visits and duration, and other online behavior. Using these technologies, online retailers are expanding their sales.

Due to the library's vital role in providing relevant information, many library users tend to borrow or rent books available in the said place. Hence, issue that library users are encountering is looking for related books or documents that the library users may use,

Information filtering systems are classified as recommendation systems. They are software applications designed to assist users in making decisions while interacting with large amounts of data. They suggest items of interest to users based on their expressed preferences, either explicitly or implicitly.

Recommender systems are a type of information filtering technology that aims to provide users with information that is likely to be of interest. When the system is unable to form any relationship between users and items for which it lacks sufficient data, it is said to be experiencing a cold start. Systematic Bootstrapping, Low Interaction, and the addition of new users are the three most common reasons for the system failing to provide recommendations (Lendave, 2021).

Acknowledging the problems encountered by librarians and library users, the developers were motivated to create a book recommender system to ease the burden of both librarians and library users.

This paper will propose recommending books using recommendation systems. We will develop a system that learns user preferences and generates a list of books the user will most likely enjoy reading.

Theoretical Framework

To define the study proposed by the proponents, there are different figures defined in this section. Figure 1 shows the architecture of a Content-based filtering recommendation system. This content-based filtering technique is based on the user's profile and the content of the item, it recommends based on the profile of the user and the similar items that the user might be interested to. There are three step process in implementing content-based filtering which are handled by individual components:

Content Analyzer. This component is responsible for the contents in a specific item coming from an information source. It extracts the gathered information and present the contents of the items such as documents, web pages, descriptions, books, etc. The data that is extracted from the sources are analyzed using various extraction techniques in order to modify item representation of the original information space to the target space. Content analyzer serves as the input for the profile learner and filtering component.

Profile Learner. This model constructs a user profile by gathering user preference data and tries to generalizes the said data. The generalization strategy is implemented through machine learning techniques in order to determine the user interests and differentiate the items that are liked and disliked. This component is responsible for identifying the possible likes of the user and the feedback based on the items that the system suggests.

Filtering Component. This component uses the user data profile to find the matching items related to the item list that the user may like. The recommended list of items will be presented in this component which in this study is the books or any reading material that is relevant to the user's preference. The matching is applied by calculating the cosine similarity between the prototype vector and the item vectors. This filtering technique is implemented due to the nature of the study wherein most of the data that will be analyzed is focusing on the user's profile.

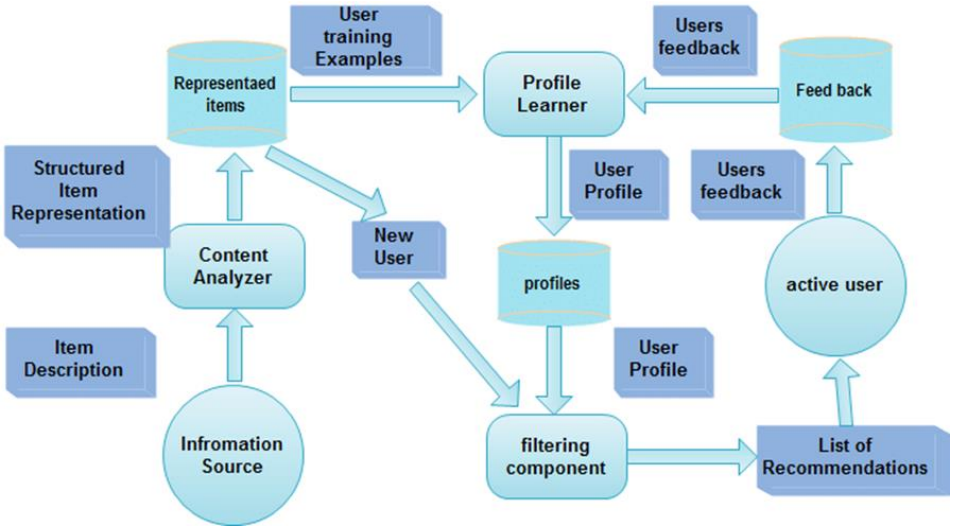

Figure 1. Content-based recommendation architecture (Mohamed et.al, 2019)

Figure 2 shows the probabilistic parameters of a Hidden Markov Model (HMM). Hidden Markov Models are a class of probabilistic graphical model to predict a sequence of hidden variables from the set of observed variables (Doraijaj, 2018). This model is used for diverse range of programs such as profile searches, and considered as the building blocks of computational sequence analysis. The x variables refer to the states which are the hidden states while the y variables represent the observable outcomes. The *a* represents the state transition probabilities moving from one hidden state to another while *b* represents the probability of moving from hidden to observable outcomes. HMM will be beneficial to the proponents' project in order to determine if there is a change of preference by the users over time.

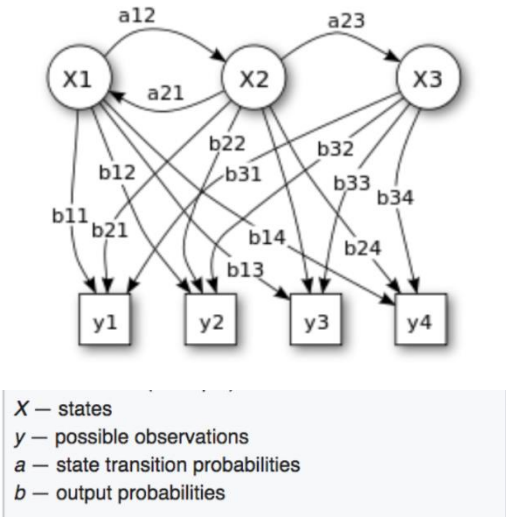

Figure 2. Hidden Markov model probabilistic parameters (Dorairaj, 2018)

Probability theory will be used as a basis of the proponents' study. Probability theory refers to the idea that every possible outcome from a sample space is assigned a numerical value of 0 and 1, known as probability measure, and a set of outcomes known as the sample space. The probability is used to make predictions in machine learning, it can be used for a model and data to make predictions based on the trained model (Agrawal, 2021). Probability theory will help in the study in order to predict the users' preference and recommend the suitable books that they might like.

Conceptual Framework

Figure 3 shows the study's conceptual framework, as described by the variables that are independent and dependent. Independent variables are the comparison of the accuracy of books recommended, and the user's borrowing history. Dependent variable

is the accuracy of the system in recommending books based on content-based filtering alone and with Hidden Markov Model.

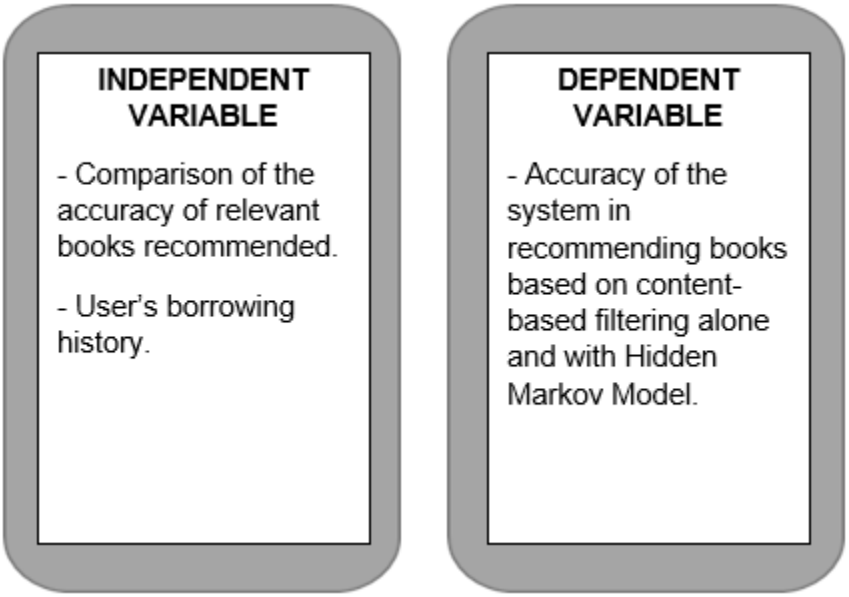

Figure 3. Conceptual Framework of the study

Statement of the Problem

This study aims to test the accuracy and develop a recommender system in an android application that will recommend books based on classical Content-based filtering with Hidden Markov Model.

The study aims to answer the following problems:

1. What is the performance and accuracy of the system in recommending books based on classical content-based filtering with Hidden Markov Model using the following methods:

1.1. Precision;

1.2. Recall;

1.3. F-score?

2. Is there a significant difference between the proposed recommender system and content-based filtering alone?

Hypothesis (H_a)

There is a significant difference between the accuracy of the proposed recommender system and content-based filtering alone.

Scope and Limitations of the Study

The proposed QR-based book recommender system will be able to keep track of all book lists, book IDs, and authors, and make recommendations based on the user's preferences and previous transactions, as well as for users who have no preferences for any items. The system should provide a friendly user interface; an average user should have no problem using it without being specially trained.

This study focused on creating a book recommender system and involves the use of the datasets obtained from Kaggle. There are 15 categories that the system will use, Fiction, Non-Fiction, Literature, Arts, Biography, Stories, Psychology, Humor, Language, History, Science, References, Medicine, Health, and Travel. This study aims to create a book recommender system using content-based filtering while utilizing QR codes.

Significance of the Study

This study will try to solve current problems in the libraries in case the book that the user wants to borrow is not available. The study will be beneficial to the following:

Librarian. The system is responsible for recommending books based on the user's interest and would lessen the load of the librarian.

Students. The book recommender system would recommend some books of interest to users based on their indicated preferences and borrowing history or in case the books that they are interested in is not available.

Future Researchers. This book recommender system will serve as a guide, all the information and data gathered in the study for the future researchers who choose this kind of topic. Also, it will serve as a reference to make a new function of the Book Recommender System.

Definition of Terms

To understand the terms stated within the study, the following terms are defined operationally:

Book recommender system. This is the recommender system the researcher aims to develop in this paper.

Bookworm. This is used to define person who is unusually devoted to reading and study

Content-based filtering. This is the technique used to help the system to recommend books.

Electronic media. This is used to define the things that can be found in a library.

Hidden Markov Model. This model is the foundation of computational sequence analysis and is used in a variety of programs such as profile searches.

Librarian. This is used to define the person who administers or assists in a library

Library. This is used as a place where the program should be used.

Library user. This is used to define person who visits the library with the purpose of exploiting its resources to satisfy his information need.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter contains the related literature that is relevant to the study. It will serve as a basis for the researchers throughout the research.

Library and its Classification

According to Estabrook et.al (2018), a library is a collection of books used for reading or study, and the building or room in which such a collection is kept. In this day and age, a library should strive to keep up with the digital world by utilizing technological advancements. Users of today, especially students of the younger generation, have access to information with just a single click.

Young and Joudrey (2020) stated that Classification is the use of a system of notations or symbols to categorize the contents of resources. It is used to arrange the shelfs logically (i.e., call numbers) and can helpful in searching the catalog. Classification refers to the “the placing of subjects into categories.” It is the process of determining where a resource fits within the categories, and assigning the resource to the closest approximates.

Library Classification is the process of arranging, grouping, and organizing books and other library materials on shelves according to their subject. Some of the known classification systems are the Library of Congress Classification (LCC), the Dewey Decimal Classification (DDC), the Bliss Bibliographic Classification (BC), the Universal Decimal Classification (UDC), Cutter Expansive Classification, and the Colon

Classification (CC). DDC and LLC are the most popular ones among them. Most of the libraries in the Philippines especially in schools and universities uses the Dewey Decimal System (DDC) so this system will be useful in the proposed research. As the researchers tackle through research studies and literatures related to the study, the researchers found out that there are not a lot of schools implementing the QR Code system in libraries.

QR code in Libraries

QR codes are two-dimensional images that, when scanned by a smartphone's camera, prompt the smartphone to open a webpage or display an image, video, or text (Kumar, Chikkamanju and Mamtha 2014). Naik et al. (2015) mentioned that with the help of the built-in QR scanners on smart phones, QR codes make it much easier to access information.

The QR code was first released in 1994 by DENSO WAVE (then a division of DENSO CORPORATION). The QR code's name refers for "quick response," representing the code's development concept, which prioritized high-speed reading. DENSO WAVE's development team embarked on the creation of a new two-dimensional code out of a genuine desire to meet the needs of users (Denso Wave, n.d.).

According to the study conducted by Durak et al. (2016), while QR Codes are easier to use, they cannot be used without technological necessities, so the use of QR Codes will increase in tandem with the increase in related technologies such as smartphones and tablets. With the spread of smartphones and tablet PCs, they may soon be the first choice for surfing and scanning quick response codes with internet access.

Parabhoi and Dhar (2017) Libraries are now being very challenged by the development of various technologies. In this present era, new technology like QR codes demands changes in information handling in the library. The user would have easy access to the most current and necessary information related to the library by using a QR code. So, to make effective use of QR codes among the user community, library professionals must organize user awareness programs, orientation programs.

Kadli (2020) conducted a study to discuss the QR Code, a new affordable emerging technology based on mobile devices, and its applications. He also discussed the primary function of the QR code and how it expedites the delivery of library services. Though QR codes and smartphones in India began relatively late, they are now becoming an integral part of daily life and accelerating technology adoption.

Gopale (2019) researched intending to identify practical QR Code applications in libraries. He also discusses the potential implementation of QR codes in library services, outlining the benefits and drawbacks. Hand trend access resources with a simple scan of a code are referred to as a treasure house of knowledge due to the high quality and quantity of information resources freely available to the academic user community.

History of Recommendation Systems

According to Muvi.com (2021), the Xerox Palo Alto Research Center created the first recommendation engine in 1992. It was made as part of the Tapestry experimental mail system. This recommendation engine's major goal was to allow users to rate messages and documents. This method proved to be effective and yielded excellent results for Tapestry. It was then improved to do more complicated tasks including filtering, retrieving, and browsing e-documents.

Isinkaye, Folajimi, and Ojokoh (2015) mentioned in their paper that there is a need to filter, select, and effectively deliver relevant information on the Internet, where the variety of options is overwhelming, in order to ease the problem of information overload, which has created a potential problem for many Internet users. Recommender systems increase the scope of personalized information extraction on the Internet.

Content-based Filtering and its Advantages and Disadvantages

Different information filtration techniques have been developed for the past few years, it has been used to design recommender systems to eliminate possible problems that may occur such as information overload and cognitive stress, these techniques provide accurate results to help the users on what books they might need (Ali, Khusro, Ullah, 2016). Ali, Khusro, Ullah (2016) stated that there are a lot of recommender systems that are out nowadays but none of them focuses on books and its contents.

Content-based filtering is a type of recommender system that attempts to guess what a user may like based on the user's activity. It makes recommendation by using keywords assigned to a database and matching them to the user's profile. (Upwork, 2021)

A lot of systems whether in filtering techniques or other fields has its advantages and drawbacks. The advantages and disadvantages of content-based filtering is listed below. (Mohamed et.al, 2019)

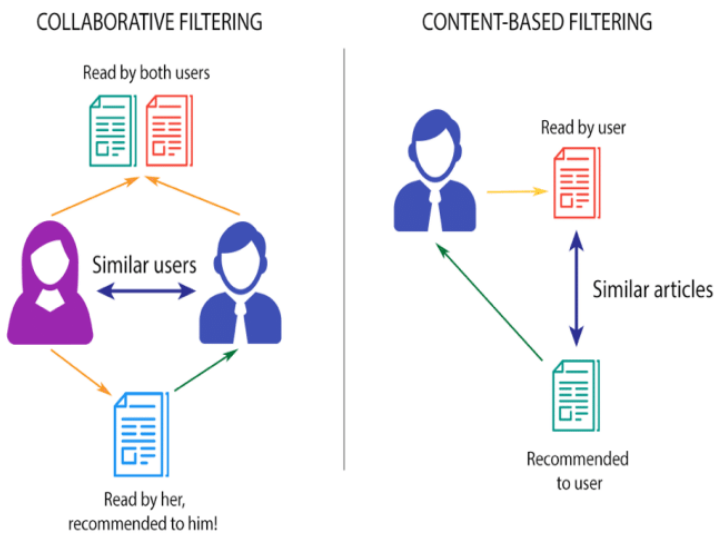

Figure 4. Content-based filtering and Collaborative filtering recommendation

One of the advantages of a content-based filtering recommendation system that it does not need the user's data to recommend items. Another is the system has the ability to recommend new items to the users based on the similarity on the items specifications, in this research, it is the book or other reading material that is related to the previous searched item.

Some drawbacks of this filtering system that it demands for analysis and detect all item features to create a list of recommendation. Another drawback is that the system does not depend on the user's rate of the item so that the evaluation of the item is not included.

Content-based filtering analyze documents or preferences given by the user, and builds a model from the said data. This technique will make use of the user's possible interests and attempt to match it to the user's profile based on the objects that will be recommended. (Reddy et.al, 2018) In the study of Reddy et.al in 2018, the researchers used content-based filtering in recommending movies to users based on the similarity of the genres that the user might be interested into. The dataset that is used for this study is

divided into two sections, one section includes the list of movies along with the genre, while the other section contains the ratings of the said movies, ranging from 1-5, with 5 being the highest. After the matrix from all the movies are generated, a similarity measure is used by computing the least distance between the user under consideration and the others. The values with the least deviation are the one recommended by the system based on the user's preferences. The implementation of machine learning in recommender systems makes recommendation more intelligent compared to the simple recommendation systems wherein few parameters are considered in recommendation.

Javed et.al in 2021 stated the advantages of a content-based recommender system which includes transparency, independence and recommendation for unclassified entities. Content-based recommendation generates recommendation based on the interaction by user either explicitly or interface tapping. From the gathered data from the user, the profile will be created to create recommendations based on the user's interaction. The researchers stated that content-based is better than collaborative because it is easy to implement and provides independence from the other users.

Other Filtering techniques

In 2021, a study conducted by Jhamat, Mustafa, Arshad, and Abbas aim to develop a hybrid book recommendation system with implicit feedback that has a purpose to compare its ability to predict user ratings in an e-book application to traditional recommender systems. Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) are used to calculate the performance of recommender systems. The results suggest that combining the Collaborative Filtering and Content-Based approaches in a hybrid model, in which the strengths of one technique balance the shortcomings of the other, improves

performance. According to the researchers, future research should look at these varied techniques by deriving recommendations and putting them into practice. A/B testing to evaluate which model performs better would bring additional validity to the study and allow for new user-behavior insights. If one were to put production, a mixed hybrid strategy may be examined. Improvements and fine-tuning of the Recommender System might help to enhance the ratings' performance.

Rana's study from 2019 shows that Collaborative Filtering (CF) is used by recommender systems to generate lists of items that are similar to the buyer's preferences. Collaborative filtering is based on the assumption that if a user has rated two books, then the other book can be recommended to a user who has read one of these books. Due to scalability, sparsity, and cold start issues, CF has difficulty providing accurate recommendations so they use this recommender with Jaccard Similarity (JS) to provide more accurate recommendations. JS is based on an index derived from a pair of books. It is calculated by dividing the number of common users (those who have rated both books) by the total number of users who have rated both books individually. The greater the number of common users, the higher the JS Index and thus better recommendations. Books with a high JS index (those that are more recommended) will appear at the top of the recommended books list.

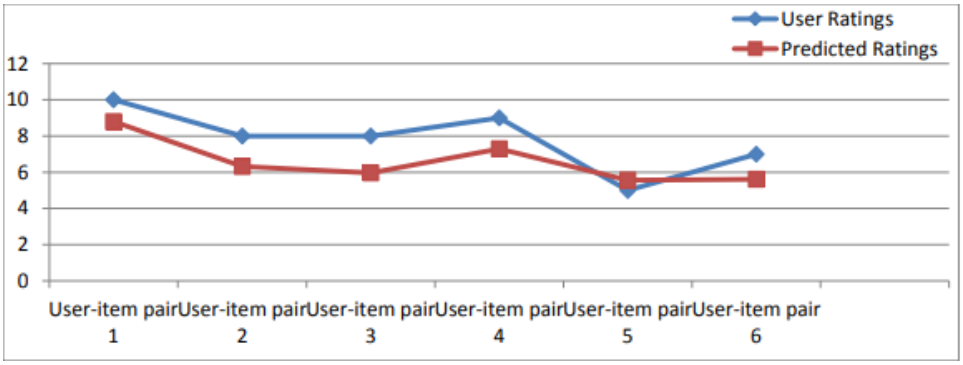

Figure 5. Comparison of Ratings and Predictions for user-item pairs

Konstantin and Nussipbekov also used a collaborative filtering method based on Pearson correlation coefficient in their paper last 2015. Collaborative filtering is a method of making product recommendations for users based on a large amount of data about their preferences. The idea behind collaborative filtering is that if two users have similar preferences for one item, they are more likely to have similar opinions on other items than with some random user. There are various types of realizations in collaborative filtering. One of them is based on neighborhood-based algorithms that determine how similar two items are. They use the Pearson correlation function to calculate similarity in their project.

Book Recommender Systems

Wang and Hou developed a collaborative filtering and interest-based book recommendation algorithm in 2021. The interest degree uses the basic attributes of the book as a measurement index, and collaborative filtering uses cosine similarity for analysis. It was concluded that the method proposed in their paper has a good convergence result after analyzing the statistical data of the library of Wuxi Vocational College of Science and Technology for many years, using MAE and RMSE two measurement indicators for experimental analysis. The next step is to improve the collaborative filtering algorithm while also improving the measurement indicators in order to achieve better convergence results.

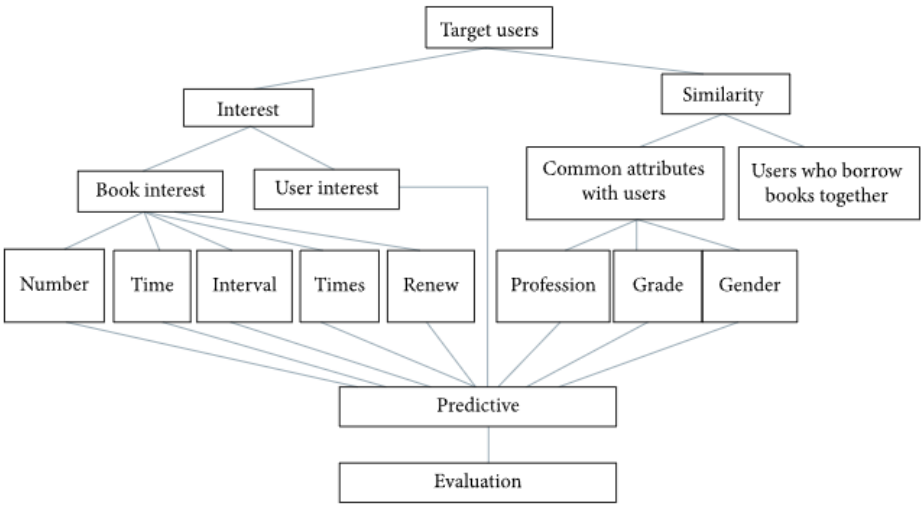

Figure 6. Recommended elements for target users

In 2018 Okon, Oke and Asagba (2018), developed an improved book recommender system using a collaborative filtering algorithm. The type of CF algorithm used by their system is the user-based collaborative filtering algorithm, it involves the process of recommending learning resources to users using previous ratings from other learners. The user profile describes a database containing user personal details, implicit and explicit ratings and list of recommendations provided based on user's similarity with other users. They introduce an efficient quick sort algorithm and adjusted cosine algorithm to improve the recommender systems.

Hidden Markov Model

Verma (2021) stated that Hidden Markov Model (HMM) is a probabilistic model that is used to explain or derive the probabilistic characteristic of any random process that is also used in machine learning. This model describes the evolution of observable events

depending on the internal factors. In this study, the HMM is responsible for storing the probability distribution of user in the latent states in the preference of user. Knowing the user's preference in books in past state would be beneficial in case the user enters the future state.

Eskandian and Mobasher (2018) used a recommendation framework based on Hidden Markov Model to detect the changes in recommendation system. The study used change point detection to identify the change points in order to target the appropriate target recommendations for the user. There are two approaches in the framework, first is the approach in change points in pre-filtering step to segment user profile and feed the segmented data into non-sequential recommendation. Second, the emission probabilities from the HMM are used to generate recommendations based on the representation of items in probability distribution on the hidden states of HMM. The method used by the proponents shows the accuracy of the results in change point detection and recommendations. Figure 7 shows the performance of different models that shows most of them are not performing well compared to the sequential models. The Sequenced-based Matrix Factorization Recommendation (SMF) with 10 hidden states leads the way and outperforming all the models. The results show the importance of change point detection and effectiveness of the method for recommendation. For the future development of this study, the researchers aim to implement the use of change point detection to other recommendation models and to use a different dataset for the study.

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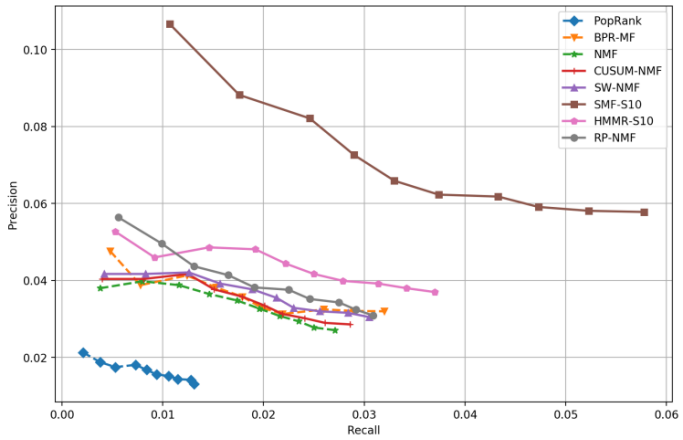

Figure 7. Comparison of Models based on Precision-Recall curves

In order to track changes in users' preferences over time, Aghdam (2019) introduces a unique hierarchical hidden Markov model that models users' latent context. The suggested technique models the user as a hidden Markov process using the user-selected items and takes into account the user's present context as a hidden variable. Using a hidden Markov model on the information gathered from the user's feedback sequences, the latent contexts are automatically discovered for each user. The experiments' findings on the test data sets demonstrate that the suggested model performs better than alternative approaches.

Aghdam, Hariri, Mobasher and Burke (2015) present a hierarchical hidden Markov model for recording changes in user preferences in this work. With the help of a user's feedback sequence on objects, they represent the user as a hierarchical hidden Markov process, with the user's current context acting as a hidden variable. The model is used to determine the highest likelihood sequence of contextual state transitions for a specific user and to forecast the probability distribution for the context of the next action. After that, suggestions are made using the anticipated context. The test findings, which were based

on music playlist data from Last.fm, show that this methodology performs much better than baseline methods in terms of accuracy and variety.

Synthesis of the Reviewed Literature and Studies

Libraries have played a huge part in every user since then, and it does not seem to slow down anytime soon. The availability of some books may be an issue for some users especially when the demand of the said book is high, so alternative resources will be beneficial for the user, the book recommender system helps in finding an alternative book that is as relevant to the book that is not available. These related literature and studies have been vital in the formulation of this research.

Book recommender systems have been emerging in the past years due to the convenience it gives to the user when their books that they want to read is not available and in need of possible replacement. The study of Oke and Asagba (2018) uses collaborative filtering algorithm in formulating their book recommender system, it is based on the ratings from the other users. The problem with this is that a collaborative filtering algorithm cannot solve a cold start problem because it depends on other users.

Content-based filtering is a type of recommender system that guess the item that the user may like based on their activity. (Upwork, 2021) The researchers have concluded that this filtering technique will be the best fit in implementing a book recommender system since it will mostly depend on the user's activity. Since most recommender systems might be static in recommending books for the users, the study of Eskandian and Mobasher in 2018 proposes a model that can detect the change of the preference of the users over time, for the future development of their study they aim to implement the change point

detection to another model, with that the proponents will try to implement the change point detection using hidden Markov model to content-based filtering.

According to the study of Javed et.al in 2021, content-based filtering provides independence and transparency to make the results of the recommendations to be highly relevant. Since the proponents are focusing on the changes in user's preference over time, content-based will be appropriate since the study since it does not require information from the other users and provides recommendation based on the user alone. The proponents are going to use content-based with HMM because there are not enough researches that implementing HMM with change point detection along with content-based to detect the change of preference of users. These conclude that the researchers will be implementing the Hidden Markov Model aside from content-based filtering alone to store the preference of a user in a particular or past state and use the stored information to recommend users when they entered the future state.

Chapter 3

METHODOLOGY

Research Design

The researchers used an experimental design for this research entitled “QR based Book Recommender System using Content-based Filtering with Hidden Markov Model”. The researchers make used of the experimental design to test the accuracy of the system to recommend books based on the user’s preference or based on the user’s past history in borrowing books using content-based filtering alone and with Hidden Markov Model.

Through this design, the researchers would be able to control the factors that can contribute to the result of the experiment. The design is used in order to test the hypothesis to determine the relationship between the dependent and independent variables to reach an acceptable conclusion. Experimental design is applied into this research to maintain accuracy of the results that the system aims in recommending books to the user and to avoid potential bias.

Sources of Data

The data are facts or information used to discuss or determine the answer to a research question. The dataset from Kaggle is the study's data source. This website’s collection of books, documents, and other printed materials, and user’s record of borrowing history can serve as a data source for the researcher.

Instrument

Researchers will use React Native as the platform for the system's functionality and interface in this research. The database that is chosen will be SQLite database based on its compatibility with React Native and will be used to store user's and book's data. A book dataset and user's record will be obtained and utilized that is available online at Kaggle website. React Native libraries will be used for the feature of the app that generates and scans QR codes for books.

Experimental design will be used to test the hypothesis. Testing will be utilized to measure the performance of the proposed Recommender System in QR-Library System. Precision, recall, and F1 score are the metrics to be used in testing the accuracy of the system. And, Mcnemar testing will be utilized to assess the hypothesis of the study.

Data Generation / Gathering Procedure

The researchers will identify the accuracy of book recommender system and the following are the data gathering procedure with content-based filtering alone and with Hidden Markov model.

The procedure to test the accuracy of the system with content-based filtering alone:

1. The researchers compiled a list of books from the dataset and insert into database, and generate QR codes for each book.
2. The proponents will search for the books they might want to read and will display the possible related books based from their borrowing history.
3. The researchers used content-based filtering for the recommendation of books.

The procedure to test the accuracy of the system with content-based filtering and hidden Markov model with cold-start solution:

1. The researchers compiled a list of books from the dataset and insert into database, and generate QR codes for each book.
2. The researchers will sign up for an account if they do not have one and choose their field of interest.
3. The proponents will search for the books they might want to read and will display the possible related books based from their borrowing history.
4. The researchers used content-based filtering with the application of Hidden Markov Model for the recommendation of the books.

Ethical Considerations

For the research, the following ethical principles will be implemented:

1. The researchers' main priority for this study is participant consent and approval.

2. The participants' dignity and well-being will be respected at all times.

3. Throughout the study, the gathered data will be kept confidential and will only be used for the purpose of this study.

Statistical Data Analysis

The aim of the study is to determine the accuracy of the content-based filtering system in recommending books or printed materials that is relevant to the user with

content-based filtering alone and with Hidden Markov Model. The proponents used the following formulas to calculate the accuracy of the recommender system.

Precision, Recall and F1 Score

The precision and recall formulas are used to evaluate the performance of classification or information retrieval systems. Precision is the fraction of relevant instances among the retrieved data. Recall is the fraction of retrieved instances among relevant data. After getting the precision and recall of the items, F1 score is computed to combine the precision and recall metrics into single metrics.

$$\begin{aligned}\text{precision} &= \frac{\text{tp}}{\text{tp} + \text{fp}} \\ &= \frac{\text{retrieved and relevant documents}}{\text{all retrieved documents}}\end{aligned}$$

$$\begin{aligned}\text{recall} &= \frac{\text{tp}}{\text{tp} + \text{fn}} \\ &= \frac{\text{retrieved and relevant documents}}{\text{all relevant documents}}\end{aligned}$$

Where **tp** = true positive, the number of instances which are relevant

fp = false positive, the number of instances that are not relevant but the model incorrectly accepted as relevant

fn = false negative, the number of instances that are not relevant are accepted as not relevant

$$F1\ score = 2 * \frac{Precision\ * Recall}{Precision + Recall}$$

A model will obtain high F1 score if both Precision and Recall are high, will obtain low F1 score if both Precision and Recall are low, and will obtain a medium F1 score if one of Precision or Recall is low and the other is high.

McNemar Test

The researchers will use the McNemar test to find if there is a significant difference between the proposed recommender system and content-based filtering alone. McNemar test is used to compare the predictive accuracy of two models. It is based on a 2 by 2 contingency table of two model predictions.

$$\chi^2 = \frac{(b-c)^2}{b+c}$$

Where:

b, c – the cells that will be evaluated

Software Development

The proponents adapted the waterfall model due to the relevance of the model to the overall architecture of the proposed system. Requirements phase is responsible for defining all the possible requirements for the system and analyzing the gathered

requirements if they are fit to the system that is being developed. Design phase studies the output from the first phase and helps in specifying the overall system architecture. With the acquired inputs from the system design, the system is implemented in small units in the Implementation stage. Each unit is developed and tested in the following stage, the Testing stage. In case that there are some issues detected in the system, the Maintenance phase comes place wherein the issues will be fixed and to deliver the changes in the environment through patches.

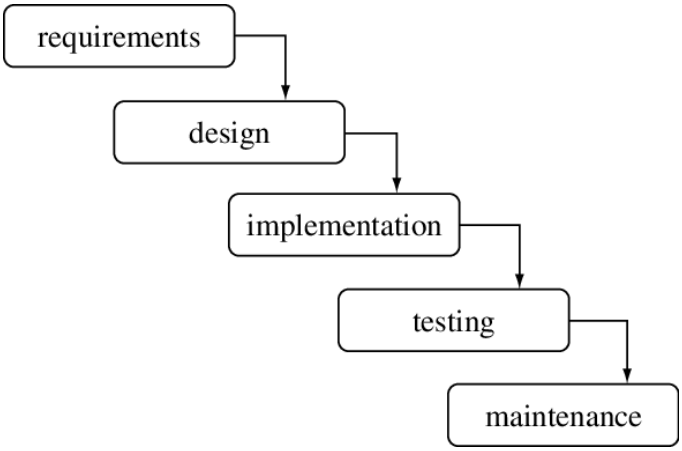

Figure 8. Waterfall model

Kaggle datasets will be collected by proponents and utilized in the experiment. React Native will be installed, together with Android Studio as an emulator as it is the main tool to develop the proposed mobile application. A compatible database will be installed for the purpose of storing books and user datasets. The design phase analyzes the results of the preceding phase and aids in defining the system architecture as a whole. Proponents will design the recommender system in detail, the application, the database, and how the data will be processed and used throughout the development once the

requirements phase is completed. Additionally, it is anticipated that this application will be created to be user-friendly. Figure 9 shows the example screen layout that would be expected. It has simple and straightforward design, icons and labels that can help the user to use it with ease. Javascript Libraries will be utilized for the feature of Qr-scanner. After the developments are implemented, testing will follow.

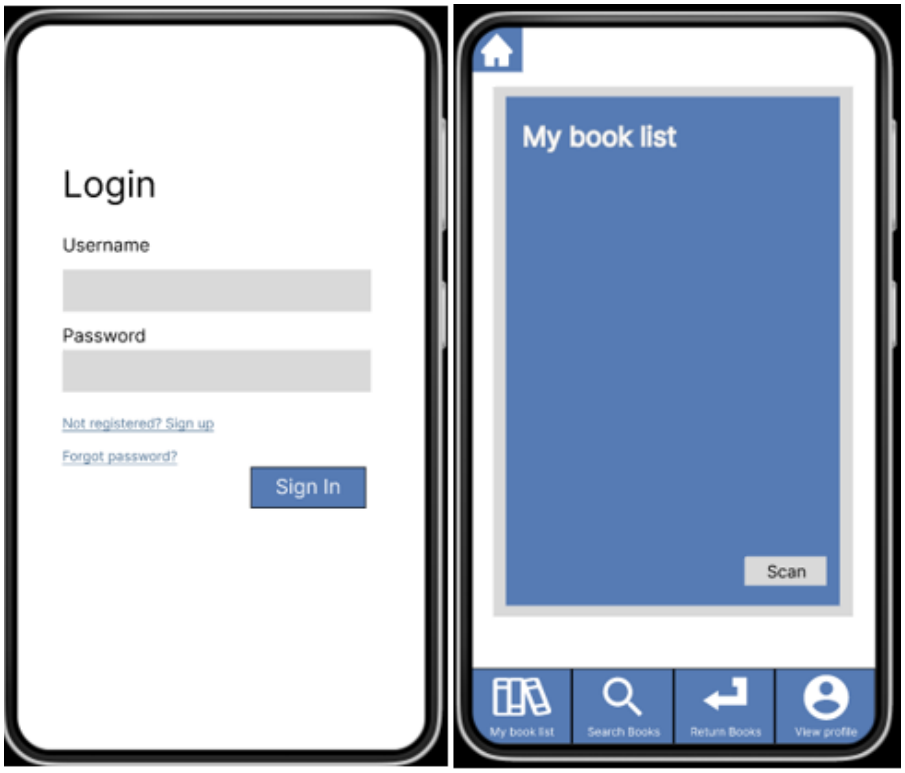

Figure 9. Sample screen layout

System Architecture

Figure 10 shows the whole system architecture of the study. For issuing of books, first, users will browse through the search function and add the books they want to borrow in their list. The recommender system (content-based filtering) will provide recommendations and rank it according to its relevance to the user's preference. Then, the user has the chance to choose and add to their list. To confirm the issuance of book(s)

in the list, QR-scanner will be utilized. For returning of books, first, users will open their list. And to confirm the returning of books, QR-scanner will also be utilized.

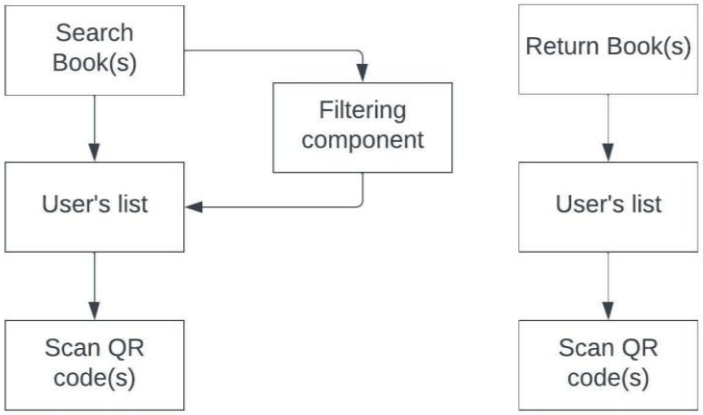

Figure 10. Overall system architecture

Figure 11 shows the system architecture of filtering component of the study. First step is to input the information, then the content analyzer arranges it and stores it into a structured item representation. After the content analyzer, the gathered information will be stored into the profile database in order to learn more about the preference of the users. After the profile learner, next step is the application of the HMM wherein the pattern of the preference of users over time will be processed through change point detection where every time the user might change preference over time, the system will know whether if the preference is static or not. The filtering component is responsible for classifying the recommendations based on the interest of the users, content-based filtering is the method used by the proponents. Then it will generate the list of recommendations and the feedback of the registered users will go to the profile learner again and will go to the profile database.

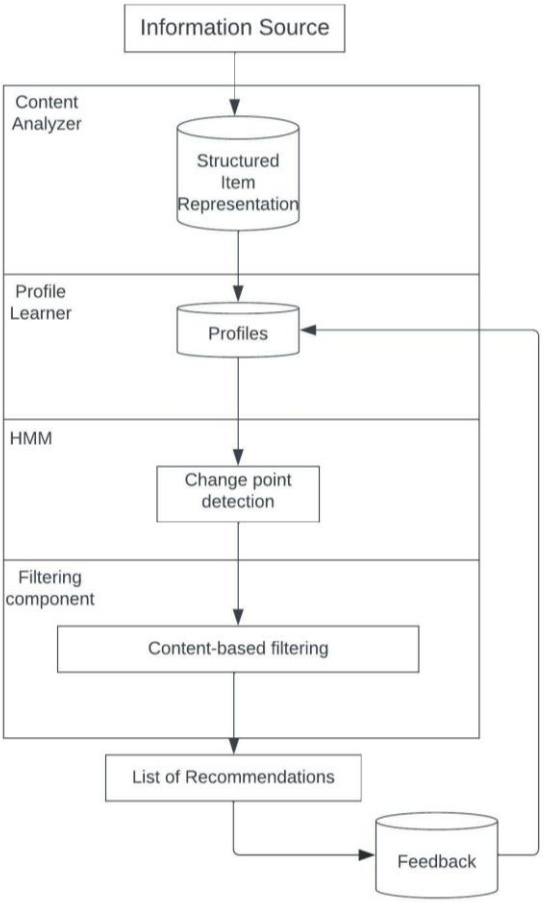

Figure 11. System architecture of filtering system with HMM
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Appendix 1:
Experiment Paper

The researchers will utilize the table below to measure the performance of Content-Based Filtering and Content-Based Filtering with Hidden Markov Model:

For both:

User	PRECISION	RECALL	F-SCORE
1			
-			
...			
N			
Average			

Table for computing the accuracy of each filtering:

Average Precision	
Average Recall	
Average F-score	
Accuracy	

Table for McNemar Test:

		Content-based with HMM	
		Correct	Wrong
Content-based	Correct		
	Wrong		
	Result		

Appendix 2

Curriculum Vitae

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